Assignment 1

Explore following 4 websites and answer the questions

- **bitcoin.org**
- Blockchain.com
- https://coinmarketcap.com/
- https://www.investopedia.com/

One-Word Answerable Questions from bitcoin.org

Bitcoin Basics

1. Bitcoin was introduced by whom? **Satoshi Nakamoto**
2. Bitcoin transactions are recorded on which data structure? **Blockchain**
3. Bitcoin uses which consensus mechanism? **Proof-of-Work**
4. The smallest unit of Bitcoin is called? **Satoshi**
5. Bitcoin is a **peer-to-peer** type of what system? **Payment**

Wallets

6. A wallet stores the user's **Private** key.
7. The recovery phrase used in wallets is called a **Seed** phrase.
8. Hardware wallets store keys **Offline** .
9. Bitcoin addresses are derived from **Public** keys.
10. What is required to sign a transaction? **Private**

Security

11. Bitcoin prevents double-spending using **Blockchain** .
12. Bitcoin is secured using **Asymmetric** cryptography.
13. To run a Bitcoin node, the software required is Bitcoin **Core** .
14. The process of verifying transactions is called **Mining** .
15. The Bitcoin network is described as **Decentralized**.

Mining

16. Miners solve **Cryptographic** puzzles to add blocks.
17. Mining rewards a block with newly created **Bitcoins** .
18. The block time target for Bitcoin is **10** minutes.
19. Mining difficulty adjusts every **2016** block.
20. Miners compete to find a valid block **Hash**.

Blockchain Structure

21. Each block contains a **Merkle** root.
22. Blocks are linked using the previous block's **Hash** .
23. A Bitcoin transaction contains inputs and **outputs** .
24. Unspent transaction outputs are called **UTXOs** .
25. The first-ever Bitcoin block is called the **Genesis** block.

Network

26. A full node stores the entire blockchain **Ledger** .

27. Bitcoin uses which protocol for communication? **P2P**

28. A light wallet uses **SPV** mode (SPV/full).

29. Nodes relay information across a **Peer-to-peer** network.

30. The maximum Bitcoin supply is **21** million.

Task B1: Explore a Block

Note the following from the **latest block**:

- Block height: **961,126**
- Block hash: 00000000000000000001473ebce7be38e82f9c332a0cafdbed00310db15e7ef9
- Timestamp:

Aug 05, 2026, 12:33:42 PM

- Number of transactions: **4,175**
- Miner/pool name: **Unknown**

Task B2: Explore a Transaction

Open any one Bitcoin transaction.

10. Write the TXID of the transaction. fe8a64cd5048f4a9d1aa384d58cd98260e29bcaf3c6716637e8cb66498437e33

11. Number of inputs: 1

12. Number of outputs: 2

13. Transaction fee: **60.000 sat/vByte**

14. How many confirmations does it have? 1

Task B3: Explore an Address

15. Search for any Bitcoin address shown in the block. bc1qwqdg6squsna38e46795at95yu9atm8azzmyvckulcc7kytlcckxswvvzej

16. Record:

- Total BTC received: **3374747.40188348 BTC**
- Total BTC sent: **3374717.88467858 BTC**
- Current balance: 29.51720490
  What is the role of a full node? A **full node** stores the complete blockchain, verifies all transactions and blocks independently, and helps secure and decentralize the Bitcoin network.

List any two developer resources available on bitcoin.org.

- Documentaries
- Podcasts

Explore : https://coinmarketcap.com/

Task C1: Market Information

- Current price of BTC: **$64,021.96**
- 24-hour % change **: 0.59%**
- Market capitalization: **$1.28T**
- Circulating supply: **20.06M BTC**
- Total/maximum supply: **21M BTC**

Task C2: Historical Data

- What was the price of Bitcoin one year ago? **$64,006.24**
- Note the highest and lowest prices in the last 30 days**. $ 66,855.57**

Explore https://www.investopedia.com/

Why is Bitcoin considered volatile?
Bitcoin is considered volatile because its price can change rapidly due to market demand, investor sentiment, government regulations, macroeconomic events, and relatively limited market liquidity compared with traditional financial assets.

List two investment risks mentioned.

- **Fraud risk**: Even with the security measures inherent within a blockchain, there are still opportunities for fraudulent activity.
- **Regulatory risk**: Laws governing cryptocurrencies continue to evolve. Changes to regulations, taxation, or trading rules could affect Bitcoin's adoption, availability, and market value.

🔹 Part E: Reflection & Analysis

Which website gave you the most clear understanding? Why?
Bitcoin.org gave the clearest understanding because it explains Bitcoin concepts, wallets, mining, security, and blockchain technology in simple language suitable for beginners.

In one paragraph, explain how blockchain ensures trust without a central authority.

Blockchain ensures trust without a central authority by storing transactions in a distributed ledger that is shared among thousands of computers (nodes). Every transaction is verified using cryptographic techniques and a consensus mechanism (Proof of Work). Once a block is added to the blockchain, altering it would require changing every subsequent block across most of the network, making fraud extremely difficult. Because every participant can independently verify transactions, users do not need to trust a bank or any central organization to maintain the integrity of the system.